

softrobot-electropneu-ctrlsys

Control elements for pneumatic actuated soft robots. Including microcontroller controller firmware, controller PCBs, control board use guidelines and more.

Closed-loop vacuum controller

This PlatformIO firmware regulates a negative-pressure pneumatic circuit with an ESP32 DevKit V1, a vacuum-pump motor, a solenoid valve, an ADS1115 ADC, and a 5 V XGZP6847A pressure sensor. It is set up for the installed **XGZP6847A100KPGPN** variant: its nominal 0.5–4.5 V output represents –100 to +100 kPa gauge pressure. The controller only commands negative pressure.

Hardware and signal path

Function	Connection	Firmware behavior
Pump motor PWM	ESP32 GPIO 2	0 is off; 255 is full duty cycle.
Solenoid valve	ESP32 GPIO 5	HIGH opens the pump-to-gripper path; LOW de-energizes the valve and vents the gripper.
Target-reached LED1 (green)	ESP32 GPIO 12 through 220 Ω to GND	On only when serial closed-loop pressure is within the configured deadband.
Active LED2	ESP32 GPIO 13 through 220 Ω to GND	On when firmware commands nonzero pump PWM or energizes the valve.
Mode switch: manual full	ESP32 GPIO 26 with 1 kΩ pulldown	3.3 V selects pump PWM 255 and valve open.
Mode switch: serial control	ESP32 GPIO 27 with 1 kΩ pulldown	3.3 V enables serial closed-loop control.
ADS1115 SDA	ESP32 GPIO 21 through the 3.3 V/5 V I2C level shifter	Initialized with Wire.begin(21, 22) .
ADS1115 SCL	ESP32 GPIO 22 through the 3.3 V/5 V I2C level shifter	Initialized with Wire.begin(21, 22) .
XGZP6847A output	ADS1115 A0	Read as a 5 V-domain analog signal; it never connects directly to the ESP32.

The ADC uses its ±6.144 V range (**GAIN_TWOTHIRDS**), so the sensor's 0.5–4.5 V signal cannot clip. After field zeroing, the nominal conversion is:

$$\text{pressure_kPa} = 50 \times (\text{sensor_volts} - \text{zero_volts})$$

ZERO removes the atmospheric offset; it does not replace a calibrated span check against a reference gauge.

ADS1115 ADDR: temporary test override

For reliable normal operation, wire ADS1115 **ADDR** to GND and use I2C address **0x48**. The ADS1115 samples this address-selection pin continuously, so a floating pin is not dependable.

For initial bench testing, `src/main.cpp` includes:

```
const bool ALLOW_FLOATING_ADS1115_ADDR_FOR_TESTING = true;
```

With this temporary toggle enabled, firmware first tries **0x48**, then the other legal ADS1115 addresses (**0x49**, **0x4A**, **0x4B**). It prints a warning if the ADC is found away from **0x48**; **STATUS** also reports **ADS_ADDR**. This only accommodates an accidentally stable floating pin. After wiring **ADDR** → **GND**, set this toggle to **false** and rebuild.

Installation and upload

1. Install [PlatformIO for VS Code](#) and open this repository as a PlatformIO project.
2. Confirm `platformio.ini` targets `esp32doit-devkit-v1`.
3. Build. PlatformIO installs [Adafruit ADS1X15](#) automatically; [Arduino](#), [Wire](#), and ESP32 [Preferences](#) are framework libraries.

```
pio run
```

4. Connect the ESP32 by USB and upload:

```
pio run --target upload
```

5. Open a serial monitor at **115200 baud** with line ending set to **Newline**:

```
pio device monitor --baud 115200
```

Boot starts deliberately uncalibrated. The firmware leaves the pump and valve off and rejects negative targets until **ZERO** succeeds.

Switch modes and LEDs

The SPDT mode switch is the output authority. The firmware reads both switch inputs on every loop and accepts only these complementary states:

GPIO 26	GPIO 27	Mode	Output behavior
HIGH	LOW	Manual full-drive	Pump PWM is 255 and the valve is energized, even if the ADS1115 is unavailable or faulted. Entering this mode cancels any serial target or calibration in progress. LED2 is on; LED1 is off.
LOW	HIGH	Serial closed- loop	The controller remains vented/idle until a new negative serial target is sent. Existing ADC, ZERO, target-range, and fault checks apply.
LOW	LOW	Safe off	Pump and valve are de-energized.
HIGH	HIGH	Safe off	Pump and valve are de-energized.

Returning from manual full-drive to serial mode does not restore an earlier target; send ZERO if needed, then a new negative target. In manual or safe-off mode, ZERO and numeric target commands are rejected. STATUS, tuning commands, SAVE, and DEFAULTS remain available.

LED1 is off while the controller ramps, vents, settles, calibrates, idles, is faulted, or is in manual/safe-off mode. LED2 represents **commanded drive**: it cannot prove that a disconnected, stalled, or failed motor/valve is drawing current because this board has no current-sense circuit.

Operation

With the pressure port open to atmosphere, first send:

```
ZERO
```

The controller turns both outputs off for one second, then averages 32 stable pressure samples. Do not obstruct the vent or run the pump while it calibrates.

After ZERO complete, send a negative gauge-pressure target as a bare number:

```
-20  
-65.93
```

The accepted range is -90 to 0 kPa. Values below -90 kPa clamp to -90; positive values clamp to zero and vent. Any zero representation stops control and vents:

```
0  
0.0  
-0.0
```

Serial commands

Command	Effect
ZERO	In serial mode, vent, establish atmospheric offset, and permit negative targets. Required after every reboot.
STATUS	Print controller state, mode, voltage, pressure, target, PWM, valve state, LED states, ADC address, calibration state, and tuning.
STATUS_STREAM	Toggle continuous STATUS output at 20 Hz. Send STATUS_STREAM again to stop it.
KP <value>	Set proportional gain in RAM; allowed range 0–50.
KI <value>	Set integral gain in RAM; allowed range 0–20.
KD <value>	Set derivative damping gain in RAM; allowed range 0–10.
BAND <kPa>	Set deadband in RAM; allowed range 0.1–5 kPa.
SAVE	Persist current validated tuning in ESP32 Preferences.
DEFAULTS	Restore conservative default tuning in RAM; send SAVE to retain it after reboot.

Malformed commands stop active control and vent as a precaution.

Controller behavior

The loop runs every 50 ms and averages four ADS1115 readings per update. PID control uses anti-windup, a filtered pressure-rate term, and a PWM slew limit.

- If pressure is not negative enough, the valve opens to the pump and PWM rises gradually.
- At or beyond the target deadband, the pump turns off while the valve remains on the pump-to-gripper path. The controller does not use the full-flow atmosphere vent as a corrective actuator.
- Passive venting (valve off) remains the safety/release path for **0**, faults, malformed commands, and **ZERO**.

This checkout's defaults are **KP=18**, **KI=1**, **KD=0.3**, and **BAND=0.5 kPa**. Tune them on the real pump, tubing volume, valve flow, and gripper before relying on them. In particular, confirm that valve-on/pump-off holds vacuum without overheating a non-continuous-duty valve coil.

Idle, off, and fault behavior

Yes: the firmware commands **both motor and valve off electrically** when it is idle or at atmospheric pressure:

```
setOutputs(PWM_OFF, false);
// pump:  analogWrite(GPIO 2, 0)
// valve: analogWrite(GPIO 5, 0)
```

This output path is used at boot, after **0**, after a positive target clamps to zero, after malformed input, during zeroing, on pressure/ADC faults, and for invalid mode-switch states. The D26 manual-full mode is

the explicit exception: it deliberately drives both outputs even with a sensor fault. With the stated plumbing, valve-off is also the pneumatic vent state; it does not attempt to hold vacuum. Therefore the valve coil is not intentionally kept energized while idle, avoiding prolonged activation and overheating.

This conclusion assumes a non-inverting valve driver where GPIO 5 HIGH energizes the coil, as defined by the original controller. Confirm that once with a multimeter or indicator LED before connecting a gripper.

First physical test

1. Test with the gripper disconnected or a conservative vented fixture.
2. Select serial mode (GPIO 27 HIGH) and send **STATUS**; verify **MODE=SERIAL**, **PWM=0**, **VALVE=VENT**, **LED1=OFF**, **LED2=OFF**, and **ZERO=REQUIRED**.
3. Run **ZERO** with the port open to atmosphere. If possible, compare reported voltage with a multimeter at ADS1115 A0.
4. Start with **-5**, then **-10**; verify motor PWM ramps rather than jumps, LED2 is on while the pump or valve is commanded, and LED1 turns on only after pressure reaches the configured deadband.
5. Send **0** and verify both PWM outputs and both LEDs are off before longer tests.
6. Select manual full-drive (GPIO 26 HIGH) and verify **PWM=255**, **VALVE=OPEN**, LED2 on, and LED1 off. Return to serial mode and verify it remains vented until a new negative target is sent.
7. Verify both-low and both-high switch input states keep both outputs and LEDs off.
8. Tune one variable at a time and send **SAVE** only after physical validation.

Never treat the -90 kPa software clamp as proof that the gripper, tubing, fittings, or pump are safe at that pressure.